

AN UNCONDITIONAL DENSITY BOUND ON ERDŐS–STRAUS EXCEPTIONAL PRIMES VIA THE SHARPNESS OF GATEWAY $A = 7$

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ABSTRACT. Let $E(X)$ denote the number of primes $p \leq X$ for which the Erdős–Straus equation $4/p = 1/x + 1/y + 1/z$ has no solution in positive integers and which are not covered by any of the known algebraic or gateway decompositions in the framework of [1]. We prove the first unconditional quantitative bound:

$$E(X) = O\left(\frac{X}{(\log X)^{3/2}}\right),$$

improving the trivial bound $O(X/\log X)$ by a factor of $(\log X)^{1/2}$. The proof uses the *sharpness* of gateway $A = 7$ (every Case B QR7 prime with an NQR prime factor in $N_7 = (p + 7)/4$ is solved by $A = 7$, proved unconditionally in [2]) to recast the exceptional set as a shifted-prime sequence sifted by the quadratic non-residues modulo 7, a sieve problem of dimension $\kappa = \frac{1}{2}$. A half-dimensional Selberg sieve with a Bombieri–Vinogradov level of distribution then gives the bound; only an upper bound is sought, so the parity obstruction does not intervene. We also characterise exactly which A values can be “sharp” in this sense, confirm that $A = 7$ is the unique sharp gateway, and describe the character-sum barrier that prevents extending the method to give $E(X) = O(1)$.

1. INTRODUCTION

The Erdős–Straus conjecture (ES) asserts that for every integer $n \geq 2$ there exist positive integers x, y, z satisfying

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

It suffices to verify the conjecture for prime values of n . Write $E(X)$ for the number of prime counter-examples up to X . Elementary congruence arguments [1] reduce the problem to the *Case B QR7* residue class: primes $p \equiv 1 \pmod{24}$ with $p \equiv 1, 2$, or $4 \pmod{7}$ for which the fraction $(p + 3)/4$ is Q_3 -smooth (all prime factors $\equiv 1 \pmod{3}$). These primes constitute approximately 2.4% of all primes, but they resist the standard algebraic decompositions.

The *gateway decomposition framework* [1] handles each Case B QR7 prime using candidate parameters A , primes $\equiv 3 \pmod{4}$, tried in increasing order until one succeeds. An initial implementation fixed a list of 28 values $A \in \{3, 7, 11, \dots, 239\}$ and confirmed that one of them always succeeds for all primes $p \leq 10^9$ (with $A = 239$ the worst case); Section 4 analyses exactly this 28-value list. Section 6 reports the corrected, baseline-validated verification of all 4,118,054,813 primes to 10^{11} (superseding an earlier run with an inverted Case A/Case B classifier, see Remark 6.1), searching with no fixed candidate cap: only 32 distinct values of A are ever required, the largest being $A = 359$ at $p = 3,807,728,761$. Exactly 47 primes below 10^{11} require $A \geq 199$, and only five require $A \geq 251$; there are no open cases.

Theoretically, the only prior bound was the trivial

$$E(X) \leq \pi_{\text{CaseB}}(X) = O\left(\frac{X}{\log X}\right),$$

where $\pi_{\text{CaseB}}(X)$ counts all Case B QR7 primes up to X . Our main result is:

Theorem 1.1 (Main). $E(X) = O(X/(\log X)^{3/2})$. *The implicit constant is effectively computable.*

The key ingredient is the exact characterisation of gateway $A = 7$ failure, proved in [2]:

Theorem 1.2 ($A=7$ Sharpness, [2, Theorem 5.2]). *For every Case B QR7 prime p , gateway $A = 7$ succeeds if and only if $N_7 = (p + 7)/4$ has at least one prime factor that is a quadratic non-residue modulo 7.*

This iff characterisation is the crucial bridge: it allows us to translate gateway failure into a multiplicative smoothness condition, amenable to analytic methods.

Remark 1.3. Theorem 1.2 holds for $A = 7$ and *no other* candidate A in the framework. Section 4 explains exactly why.

Organisation. Section 2 recalls notation and the gateway machinery. Section 3 proves Theorem 1.1. Section 4 characterises sharp gateways. Section 5 describes the character-sum approach and the barrier to unconditional finitude. Section 6 reports the verification results to 10^{11} . Section 7 lists open problems.

2. SETUP AND NOTATION

We follow the notation of [1, 2] throughout.

For a prime $A \equiv 3 \pmod{4}$, write Q_A for the set of quadratic residues modulo A (including 0):

$$Q_A = \{r \in \mathbb{Z}/A\mathbb{Z} : r \equiv x^2 \pmod{A} \text{ for some } x\}.$$

The non-trivial quadratic residues form a subgroup of $(\mathbb{Z}/A\mathbb{Z})^*$ of index 2 and order $(A-1)/2$.

Definition 2.1 (Q_A -smooth). A positive integer n is Q_A -smooth if every prime factor q of n satisfies $\left(\frac{q}{A}\right) = +1$.

For a Case B QR7 prime p and a candidate parameter $A \equiv 3 \pmod{4}$ with $4 \mid (p+A)$, set $N_A = (p+A)/4$. The *gateway target* is

$$t_A = -4N_A^2 \pmod{A},$$

and the *gateway set* is

$$S_A(p) = \{d \pmod{A} : d \mid N_A^2, \gcd(d, Ap) = 1\}.$$

Gateway A *succeeds* for prime p if $t_A \in S_A(p)$.

The key lemma [1, Lemma 3.1] asserts:

Lemma 2.2 (NQR Target). *For every prime $A \equiv 3 \pmod{4}$, the target t_A is a quadratic non-residue modulo A .*

This follows immediately from $(-1/A) = -1$ when $A \equiv 3 \pmod{4}$.

Theorem 2.3 (Failure Characterisation, [2, Theorem 3.2]). (i) *If N_A is Q_A -smooth, then $S_A(p) \subseteq Q_A \cup \{0\}$, so $t_A \notin S_A(p)$ and gateway A fails.*
(ii) *Conversely, if gateway A succeeds, then N_A is not Q_A -smooth.*
(iii) *Having an NQR prime factor is necessary but not sufficient for gateway success: $S_A(p)$ may contain NQR residues without containing the specific target t_A .*

From (i): $\{p : N_A \text{ is } Q_A\text{-smooth}\} \subseteq \{p : \text{gateway } A \text{ fails}\}$.

From (ii): $\{p : \text{gateway } A \text{ succeeds}\} \subseteq \{p : N_A \text{ is not } Q_A\text{-smooth}\}$.

3. THE UNCONDITIONAL DENSITY BOUND

3.1. Reduction to a smoothness count.

Lemma 3.1. $E(X) \leq \#\{p \leq X : p \text{ Case B QR7, } N_7 \text{ is } Q_7\text{-smooth}\}.$

Proof. If p is an exceptional prime ($E(X)$ counter-example), every gateway fails. In particular, gateway $A = 7$ fails. By Theorem 1.2, the exact characterisation for $A = 7$, gateway $A = 7$ fails if and only if N_7 is Q_7 -smooth. Hence every exceptional prime has N_7 Q_7 -smooth. $\square$

Remark 3.2. The inclusion is tight: if a Case B QR7 prime p has N_7 Q_7 -smooth, then gateway $A = 7$ certainly fails, but some other gateway $A \neq 7$ might still succeed for p . Thus the right-hand side of Lemma 3.1 is in general strictly larger than $E(X)$.

3.2. Heuristic motivation: the density of Q_7 -smooth integers.

Lemma 3.3 (Landau–Selberg for Chebotarev sets). *Let $\mathcal{P}$ be a set of primes with Dirichlet (Chebotarev) density $\delta \in (0, 1)$. Call n $\mathcal{P}$ -smooth if every prime factor of n lies in $\mathcal{P}$. Then*

$$\#\{n \leq X : n \text{ is } \mathcal{P}\text{-smooth}\} \sim C_{\mathcal{P}} \frac{X}{(\log X)^{1-\delta}}$$

for an explicit positive constant $C_{\mathcal{P}}$.

This is a classical consequence of the Selberg–Delange method applied to the Dirichlet series $\sum_{n \in \mathcal{P}\text{-smooth}} n^{-s} = \prod_{q \in \mathcal{P}} (1 - q^{-s})^{-1}$, which has a singularity of order δ at $s = 1$; see [5, Chapter II.5] for the full statement.

For Q_7 -smooth integers: the set of QR primes mod 7 has Chebotarev density $\delta = 1/2$ (half of all primes, by quadratic reciprocity and Dirichlet’s theorem). Lemma 3.3 gives:

$$(1) \quad \#\{n \leq X : n \text{ is } Q_7\text{-smooth}\} \sim C_7 \frac{X}{(\log X)^{1/2}}.$$

If the values $N_7 = (p+7)/4$ behaved, with respect to Q_7 -smoothness, like random integers of size $X/4$, then by (1) a proportion $\asymp (\log X)^{-1/2}$ of the $\asymp X/\log X$ admissible primes would have N_7 smooth, predicting $E(X) \asymp X/(\log X)^{3/2}$. We make this rigorous as an *upper* bound. The integer N_7 is a shifted prime, not a random integer, so the heuristic must be replaced by a sieve. Because only an upper bound is required, the parity obstruction does not arise, and a half-dimensional sieve with a Bombieri–Vinogradov level of distribution suffices.

3.3. The exceptional set as a sifted sequence. Fix the three residue classes modulo $168 = \text{lcm}(24, 7)$ determined by $p \equiv 1 \pmod{24}$ and $p \equiv 1, 2, 4 \pmod{7}$; write $\mathcal{R} \subseteq (\mathbb{Z}/168\mathbb{Z})^*$ for this set, so $|\mathcal{R}| = 3$. Define the sifting sequence

$$\mathcal{A} = \left\{ (p+7)/4 : p \leq X \text{ prime, } p \bmod 168 \in \mathcal{R} \right\},$$

and the sifting set of primes (the non-residues modulo 7)

$$\mathcal{P} = \left\{ \ell \text{ prime} : \ell \equiv 3, 5, 6 \pmod{7} \right\}.$$

Write $P(z) = \prod_{\ell \in \mathcal{P}, \ell < z} \ell$ and

$$S(\mathcal{A}, \mathcal{P}, z) = \#\{a \in \mathcal{A} : \gcd(a, P(z)) = 1\}.$$

Lemma 3.4 (Smoothness forces survival). *For every $z \geq 2$,*

$$\#\{p \leq X : p \bmod 168 \in \mathcal{R}, N_7 \text{ is } Q_7\text{-smooth}\} \leq S(\mathcal{A}, \mathcal{P}, z).$$

Proof. If $N_7 = (p+7)/4$ is Q_7 -smooth then every prime factor of N_7 is a quadratic residue modulo 7, so N_7 has no prime factor in $\mathcal{P}$; in particular $\gcd(N_7, P(z)) = 1$, and p contributes the element $a = N_7$ to the survivors. Here we use that for odd $\ell \neq 7$, $\ell \mid (p+7)/4 \iff \ell \mid (p+7)$ since $\gcd(\ell, 4) = 1$; every $\ell \in \mathcal{P}$ is such, because $2, 7 \notin \mathcal{P}$ (indeed $2 \equiv 3^2$ is a residue and $7 \equiv 0 \pmod{7}$). $\square$

Combining Lemmas 3.1 and 3.4, $E(X) \leq S(\mathcal{A}, \mathcal{P}, z)$ for every $z \geq 2$.

3.4. Local densities and the sieve dimension.

Lemma 3.5. *For squarefree $d \mid P(z)$ write $\mathcal{A}_d = \{a \in \mathcal{A} : d \mid a\}$. Then $\gcd(d, 168) = 1$ and*

$$|\mathcal{A}_d| = \frac{\omega(d)}{d} X_0 + r_d, \quad \frac{\omega(d)}{d} = \frac{1}{\phi(d)}, \quad X_0 = \frac{3}{\phi(168)} \text{Li}(X),$$

where $r_d = \sum_{i=1}^3 (\pi(X; 168d, c_i) - \text{Li}(X)/\phi(168d))$ for residues c_i supplied by the Chinese Remainder Theorem. The density ω is multiplicative with $\omega(\ell)/\ell = 1/(\ell-1)$ for $\ell \in \mathcal{P}$, and the sieve has dimension $\kappa = \frac{1}{2}$:

$$V(z) := \prod_{\substack{\ell < z \\ \ell \in \mathcal{P}}} \left(1 - \frac{\omega(\ell)}{\ell}\right) = \prod_{\substack{\ell < z \\ \ell \in \mathcal{P}}} \left(1 - \frac{1}{\ell-1}\right) \asymp (\log z)^{-1/2}.$$

Proof. For $\ell \in \mathcal{P}$ we have $\ell \nmid 168$, and as in Lemma 3.4 $\ell \mid a = (p+7)/4 \iff p \equiv -7 \pmod{\ell}$. Since $\gcd(-7, \ell) = 1$ this is one class modulo ℓ , coprime to ℓ . By the Chinese Remainder Theorem the conditions “ $p \bmod 168 \in \mathcal{R}$ and $p \equiv -7 \pmod{d}$ ” select exactly 3 classes modulo $168d$, all coprime to $168d$ (using $\gcd(d, 168) = 1$). Counting primes in these classes gives $|\mathcal{A}_d|$ with main term $3 \text{Li}(X)/\phi(168d) = \frac{1}{\phi(d)} \cdot \frac{3 \text{Li}(X)}{\phi(168)}$, i.e. $\omega(d)/d = 1/\phi(d)$, multiplicative with $\omega(\ell)/\ell = 1/(\ell-1)$.

For the dimension, $\log(1 - \frac{1}{\ell-1}) = -\frac{1}{\ell} + O(\ell^{-2})$, so

$$-\log V(z) = \sum_{\substack{\ell < z \\ \ell \in \mathcal{P}}} \frac{1}{\ell} + O(1) = \frac{1}{2} \log \log z + O(1),$$

by Mertens' theorem for arithmetic progressions: the residues 3, 5, 6 (mod 7) comprise three of the $\phi(7) = 6$ reduced classes, each contributing $\frac{1}{6} \log \log z + O(1)$ to $\sum_{\ell < z} 1/\ell$. Exponentiating gives $V(z) \asymp (\log z)^{-1/2}$. $\square$

3.5. Level of distribution.

Lemma 3.6 (Bombieri–Vinogradov input). *For every $K > 0$, with $D = X^{2/5}$,*

$$\sum_{\substack{d \leq D \\ d|P(z)}} \mu^2(d) 3^{\nu(d)} |r_d| \ll_K \frac{X}{(\log X)^K}.$$

Proof. By the triangle inequality, writing $\Delta(q, a) = \pi(X; q, a) - \text{Li}(X)/\phi(q)$,

$$|r_d| \leq \sum_{i=1}^3 |\Delta(168d, c_i)| \leq 3 \max_{(a, 168d)=1} |\Delta(168d, a)|,$$

and $168d \leq 168D \leq X^{1/2}$ for large X . Cauchy–Schwarz gives

$$\sum_{d \leq D} 3^{\nu(d)} |r_d| \leq \left(\sum_{d \leq D} \frac{9^{\nu(d)}}{\phi(d)} \cdot X \log X \right)^{1/2} \left(\sum_{d \leq D} |r_d| \right)^{1/2},$$

using the trivial bound $|r_d| \ll X \log X / \phi(d)$. The first factor is $\ll X^{1/2} (\log X)^5$ since $\sum_{d \leq D} 9^{\nu(d)} / \phi(d) \ll (\log X)^9$; the second is $\ll (X / (\log X)^{K'})^{1/2}$ for any K' by the Bombieri–Vinogradov theorem [4, Thm. 17.1], valid since $168D \leq X^{1/2}$. Choosing K' large gives the claim. $\square$

3.6. Proof of Theorem 1.1. Take $z = X^{1/5}$, so the Selberg upper-bound sieve with weights supported on $d \leq \xi = z$ has remainder over moduli $d \leq D = \xi^2 = X^{2/5}$. By Lemma 3.5 the sequence $\mathcal{A}$ satisfies the κ -dimensional hypothesis of Selberg's sieve with $\kappa = \frac{1}{2}$ and density ω (note $\omega(\ell)/\ell = 1/(\ell - 1) \leq \frac{1}{2}$ for $\ell \geq 3$), and by Lemma 3.6 it has level of distribution $D = X^{2/5} \leq X^{1/2-\epsilon}$, within Bombieri–Vinogradov range. Selberg's sieve [3, Thm. 2.2 and the κ -dimensional bound of §2.4], [4, Ch. 6] yields

$$S(\mathcal{A}, \mathcal{P}, z) \ll X_0 V(z) + \sum_{\substack{d \leq D \\ d|P(z)}} \mu^2(d) 3^{\nu(d)} |r_d|.$$

By Lemma 3.5, $V(z) \asymp (\log z)^{-1/2} \asymp (\log X)^{-1/2}$ since $\log z = \frac{1}{5} \log X$, and $X_0 \asymp X / \log X$; the remainder is $\ll_K X / (\log X)^K$ for any K by Lemma 3.6. Hence

$$S(\mathcal{A}, \mathcal{P}, z) \ll \frac{X}{\log X} \cdot \frac{1}{(\log X)^{1/2}} = \frac{X}{(\log X)^{3/2}}.$$

By Lemmas 3.1 and 3.4, $E(X) \leq S(\mathcal{A}, \mathcal{P}, z)$, proving the theorem. The implied constant is effective, as the Bombieri–Vinogradov theorem and the Selberg bound are effective for the moduli involved. $\square$

Remark 3.7 (Only an upper bound is claimed). The argument bounds $E(X)$ from above and uses no asymptotic for $S(\mathcal{A}, \mathcal{P}, z)$. A matching lower bound, or an asymptotic for $\#\{p \leq X : N_7 \text{ is } Q_7\text{-smooth}\}$, would meet the parity obstruction of sieve theory and is not needed here. The exponent $3/2$ is exactly $1 + \kappa$ with $\kappa = \frac{1}{2}$ the relative density of non-residues modulo 7; the saving $(\log X)^{-1/2}$ is intrinsic to the quadratic splitting at $A = 7$.

4. SHARP GATEWAYS AND THE UNIQUENESS OF $A = 7$

Definition 4.1 (Sharp gateway). A candidate parameter $A \equiv 3 \pmod{4}$ is *sharp* if, for every Case B QR7 prime p for which $4 \mid (p + A)$, gateway A succeeds *if and only if* N_A is not Q_A -smooth; and moreover the characterisation is *non-vacuous*: there exist Case B QR7 primes p for which N_A is not Q_A -smooth (so gateway A sometimes succeeds on Case B QR7 primes).

Remark 4.2. The non-vacuity clause excludes $A = 3$. By definition, Case B QR7 primes satisfy $(p + 3)/4$ is Q_3 -smooth, so $N_3 = (p + 3)/4$ is always Q_3 -smooth and gateway $A = 3$ always fails on Case B QR7 primes. The iff holds vacuously but gives no density improvement.

By Theorem 2.3(i)(ii), every gateway satisfies the forward implication (Q_A -smooth $\Rightarrow$ fails) and the converse direction at the level of necessary conditions (success $\Rightarrow$ not smooth). A sharp gateway additionally satisfies the reverse: not smooth $\Rightarrow$ succeeds.

The proof of Theorem 1.2 in [2] proceeds by:

- (1) Identifying the *forced* QR factor $g = 2$ of N_7 : since $p \equiv 1 \pmod{8}$, we have $8 \mid (p + 7)$, hence $2 \mid N_7$.
- (2) Observing that 2 *generates* $Q_7 = \{1, 2, 4\}$: the powers $2^0, 2^1, 2^2$ cycle through all of $Q_7 \pmod{7}$.
- (3) Showing that if $q \mid N_7$ with $\left(\frac{q}{7}\right) = -1$, then $q \cdot 2^j \pmod{7}$ hits every NQR modulo 7 as $j \in \{0, 1, 2\}$, including the target t_7 .
- (4) Verifying the integrality condition: $d = q \cdot 2^{2k}$ divides N_7^2 (since $v_2(N_7^2) = 2v_2(N_7) \geq 2$), and $\gcd(d, 7p) = 1$ by Lemma 3.2 of [1].

Theorem 4.3 (Uniqueness of $A = 7$). *Among the 28 candidate A values in the framework, $A = 7$ is the only sharp gateway.*

Proof. Sharpness requires a prime g that is *forced* to divide N_A for every Case B QR7 prime with $4 \mid (p + A)$, such that (a) g is a QR

modulo A , (b) g generates Q_A under multiplication modulo A , and (c) the guaranteed valuation suffices: $2v_g(N_A) \geq |Q_A| - 1$, so that $\{g^0, g^1, \dots, g^{2v_g}\}$ realises all of Q_A as divisor residues of N_A^2 .

Which primes can be forced. Case B QR7 fixes $p \equiv 1 \pmod{8}$ and $p \equiv 1 \pmod{3}$, but leaves $p \bmod \ell$ free for every prime $\ell \geq 5$. Hence the only primes that can divide $N_A = (p + A)/4$ for *all* such p are 2 and 3, and a direct check of $4N_A = p + A$ gives the rule

$$2 \mid N_A \iff A \equiv 7 \pmod{8}, \quad 3 \mid N_A \iff A \equiv 2 \pmod{3}.$$

A candidate with neither $A \equiv 7 \pmod{8}$ nor $A \equiv 2 \pmod{3}$ has no forced prime factor and cannot be sharp.

We check each candidate:

$A = 7$: $7 \equiv 7 \pmod{8}$, so $g = 2$ is forced; $7 \equiv 1 \pmod{3}$, so 3 is not. $(2/7) = +1$, and $\text{ord}_7(2) = 3 = |Q_7|$, so 2 generates Q_7 . The guaranteed $v_2(N_7) \geq 1$ gives $2v_2(N_7) \geq 2 = |Q_7| - 1$. **Sharp.**

$A = 11$: $11 \equiv 2 \pmod{3}$, so $g = 3$ is forced ($3 \mid N_{11}$); $11 \equiv 3 \pmod{8}$, so 2 is not. $(3/11) = +1$ (QR, since $3 \in \{1, 3, 4, 5, 9\}$). Order of 3 in $(\mathbb{Z}/11\mathbb{Z})^*$ is 5 = $|Q_{11}|$, so 3 generates Q_{11} . However, the minimum forced multiplicity is $v_3(N_{11}) \geq 1$, giving $2v_3(N_{11}) \geq 2$, but we need $\geq |Q_{11}| - 1 = 4$. For $p \equiv 1, 4 \pmod{9}$ (two-thirds of cases) one has $v_3(N_{11}) = 1$, so the required divisors 3^3 and 3^4 do not appear in N_{11}^2 and the target t_{11} need not be hit.

Concrete counterexample. Take $p = 193$ (Case B QR7: $193 \equiv 1 \pmod{24}$, $193 \equiv 4 \pmod{7}$, $(p + 3)/4 = 49 = 7^2$ is Q_3 -smooth). Then $N_{11} = (193 + 11)/4 = 51 = 3 \times 17$. The factor 17 satisfies $(17/11) = -1$ (NQR: $6 \notin Q_{11}$), so N_{11} has an NQR prime factor. Yet $t_{11} = (-4 \cdot 51^2) \bmod 11 = 2$, while $S_{11}(193) = \{1, 3, 5, 6, 7, 9, 10\}$ (all residues of divisors of $51^2 = 3^2 \cdot 17^2$ modulo 11), and $2 \notin S_{11}(193)$. Gateway $A = 11$ fails for $p = 193$ despite N_{11} having an NQR factor. **Not sharp.**

$A = 19$: $19 \equiv 3 \pmod{8}$ and $19 \equiv 1 \pmod{3}$, so *neither* 2 nor 3 is forced into N_{19} . No prime is forced, so condition for sharpness fails at the first step. (For instance $p = 193$ is Case B QR7 with $N_{19} = 53$, prime and coprime to 2 and 3.) **Not sharp.**

$A \geq 23$ (**remaining 25 candidates**): A systematic check of all 28 candidate values confirms that for every $A \in \{23, 31, 43, \dots, 239\}$, either (a) no forced prime ($A \not\equiv 7 \pmod{8}$ and $A \not\equiv 2 \pmod{3}$, e.g. $A = 43, 67, 79, \dots$), or (b) a forced prime exists but the guaranteed valuation satisfies $2v_g(N_A) < |Q_A| - 1$. For $A = 23$ ($\equiv 7 \pmod{8}$ and $\equiv 2 \pmod{3}$) both 2 and 3 are forced, but $|Q_{23}| = 11$ requires $2v_g \geq 10$ while the guaranteed valuations are 1 each. The growing

gap $|Q_A| - 1 \geq 10$ against bounded forced valuations rules out every $A \geq 23$. **None sharp.**

The computation was verified by testing all 28 candidates directly, confirming $A = 7$ is the unique potentially sharp gateway. $\square$

Remark 4.4. The special arithmetic of $A = 7$: $|Q_7| = 3$ is the smallest possible, and $|Q_7| - 1 = 2$ equals $2 \cdot \min(v_2(N_7)) = 2 \cdot 1$. No larger prime A can match this coincidence.

5. CHARACTER SUMS AND THE BARRIER TO FINITUDE

The density bound of Theorem 1.1 does not imply finiteness of E : $X/(\log X)^{3/2} \rightarrow \infty$ as $X \rightarrow \infty$. We describe the character-sum approach that would be needed to obtain $E(X) = O(1)$.

5.1. Character sum formula for gateway solutions. The number of gateway solutions for parameter A is:

$$(2) \quad \mathcal{N}_A(p) = \frac{1}{\varphi(A)} \sum_{\chi \bmod A} \bar{\chi}(t_A) \sum_{\substack{d|N_A^2 \\ \gcd(d, Ap)=1}} \chi(d).$$

Separating the trivial character χ_0 and the Legendre symbol χ_2 :

$$(3) \quad \mathcal{N}_A(p) = \frac{\tau(N_A^2) - \tau(M_A^2)}{\varphi(A)} + E_A(p),$$

where M_A is the Q_A -smooth part of N_A , and $E_A(p)$ involves only characters of order ≥ 4 .

The *main term* $\tau(N_A^2) - \tau(M_A^2)$ is positive if and only if N_A is not Q_A -smooth (i.e., $M_A < N_A$). Gateway A fails only when $\mathcal{N}_A(p) = 0$, which requires $E_A(p) \leq -(\tau(N_A^2) - \tau(M_A^2))/\varphi(A)$.

5.2. Why the character sum barrier is hard. For $A = 7$: $\varphi(7) = 6$ has characters of orders 1, 2, 3, 6. The order-3 characters are the “next-order” corrections. Bounding $|E_7(p)|$ requires nontrivial estimates for twisted divisor sums $\sum_{d|N_7^2} \chi_3(d)$.

For general A : bounding $\sum_{p \leq X} |\mathcal{N}_A(p)|^{-\epsilon}$ (which would suffice for finitude under GRH-type hypotheses) is an open problem.

5.3. What additional hypotheses give.

Theorem 5.1 (Conditional, [2, Theorem 7.2]). *Under Conjecture 7.1 of [2] (joint Q_A -smoothness equidistribution, weaker than Elliott–Halberstam), the Erdős–Straus conjecture holds for all sufficiently large primes.*

The gap between Theorem 1.1 and finiteness corresponds to the gap between:

- $O(X/(\log X)^{3/2})$ (what smoothness density gives), and
- $O(1)$ (what finiteness requires).

To close this gap unconditionally, one would need to show that the character sum error $E_A(p)$ is never large enough to cancel the main term across all 28 gateways simultaneously.

5.4. Why more gateways do not trivially improve the exponent. It is tempting to improve Theorem 1.1 by using all 28 candidates at once. The naive heuristic runs as follows. For primes $q > 239$ each q divides at most one N_{A_i} (since $N_{A_i} - N_{A_j} = (A_i - A_j)/4 \leq 58 < q$ in the 28-candidate set), and by Chebotarev the Frobenius class of q at each gateway A_i is independent. The probability that every large prime factor of N_{A_i} is QR mod A_i simultaneously for $i = 1, \dots, k$ would then decay as $(\log X)^{-k/2}$, suggesting $E(X) = O(X/(\log X)^{1+k/2})$.

This heuristic is *not rigorous*, and the obstruction is structural. The bound of Theorem 1.1 uses the *sharpness* of $A = 7$: the exceptional set is contained in $\{p : N_7 \text{ is } Q_7\text{-smooth}\}$ because for $A = 7$, gateway failure is *equivalent* to Q_7 -smoothness. For every other A the equivalence fails (Theorem 2.3(iii)): a prime can fail gateway A_i while N_{A_i} is *not* Q_{A_i} -smooth. Consequently the exceptional set is *not* contained in the joint-smoothness set $\bigcap_i \{N_{A_i} \text{ is } Q_{A_i}\text{-smooth}\}$, and bounding the latter by a Selberg sieve gives *no* upper bound on $E(X)$. Indeed $\bigcap_i \{N_{A_i} \text{ smooth}\} \subseteq E$ runs the wrong way, a lower bound on E if anything.

A genuine improvement therefore requires controlling the *character-sum* failure events $\{\mathcal{N}_{A_i}(p) = 0\}$ of (2) for $A_i \neq 7$, not merely smoothness. We record the expected truth as a conjecture, with the caveat that it is not a sieve statement:

Conjecture 5.2. $E(X) = O(X/(\log X)^{1+\delta})$ for every $\delta > 0$; conjecturally $E(X) = O(1)$.

A proof would need joint control of the divisor character sums across the candidate gateways, the analytic gap described in [2]; the smoothness method of Theorem 1.1 is intrinsically limited to the single sharp gateway $A = 7$ and the exponent $3/2$.

6. VERIFICATION TO 10^{11}

6.1. Method. The reported numbers come from the corrected, baseline-validated pipeline (`sessions/session20_corrected_10B.py` in the software repository), which:

- (1) generates primes with a CPU segmented sieve (segments of 10^8);
- (2) classifies each prime into the algebraic cases of [1] or the Case B QR7 residual class, using the corrected Case A/Case B test (Remark 6.1);
- (3) for each Case B QR7 prime, searches candidate values $A \equiv 3 \pmod{4}$ in increasing order — with no fixed upper cap, up to $A < 10^4$ — factorising N_A and testing the divisor-residue condition of Section 2, recording the first successful A ;
- (4) runs on a 16-worker CPU pool with resumable, per-segment checkpointing, validating against three independent baselines (the full A -distribution at 10^6 , the residual count at 10^7 , and the residual count and max A at 10^9) before extending the range.

An earlier GPU-accelerated implementation (CuPy, fixed candidate lists of 28 then 87 values) first located the $A = 359$ record described in Section 6.3; it has since been superseded by the pipeline above, which is the source of every number in this section.

Remark 6.1 (Data integrity). An earlier run of the 10^{10} stage (March 2026) inverted the Case A/Case B test, skipping every residual-class prime as already proven and instead gateway-searching the complementary (easy) class. The resulting figures — a residual count of 18,189,229 at 10^{10} , max $A = 251$, a 25-value A -dictionary — described the wrong set of primes and are withdrawn; they do not appear in this paper. The corrected pipeline described above reproduces the correct-classifier baselines at 10^6 , 10^7 , and 10^9 before extending the range, and every reported solution is verified by direct evaluation of $4/p = 1/x + 1/y + 1/z$.

6.2. Results to 10^{11} .

Limit	Primes	Case B QR7	Max A required
10^6	78,498	2,269	79
10^7	664,579	18,019	167
10^8	5,761,455	146,227	239
10^9	50,847,534	1,212,383	239
10^{10}	455,052,511	10,246,512	359 (see §6.3)
10^{11}	4,118,054,813	88,088,687	359 (unchanged from 10^{10})

Theorem 6.2 (Computational Verification). *The Erdős–Straus conjecture holds for all primes $p \leq 10^{11}$, with maximum gateway parameter $A \leq 359$ and zero open cases. Exactly 47 primes below 10^{11} require $A \geq 199$ (12 of them below 10^{10}), and only five require $A \geq 251$: $p = 3,807,728,761$ ($A = 359$), $p = 31,048,497,481$ ($A = 347$), $p = 6,372,847,849$ ($A = 311$), $p = 26,147,464,729$ ($A = 251$), and*

$p = 94,604,730,049$ ($A = 251$). The record $A = 359$ is attained below 10^{10} and is not exceeded anywhere in $(10^{10}, 10^{11}]$.

6.3. Anatomy of the hardest prime. The unique prime below 10^{11} requiring $A = 359$ is

$$p = 3,807,728,761.$$

Verification that p is Case B QR7. $p \equiv 1 \pmod{24}$, $p \equiv 4 \pmod{7}$, and $(p+3)/4 = 951,932,191 = 7 \times 73 \times 1,862,881$, all three factors $\equiv 1 \pmod{3}$, confirming Q_3 -smoothness.

Solution via $A = 359$. $N_{359} = (p+359)/4 = 951,932,280 = 2^3 \cdot 3 \cdot 5 \cdot 13 \cdot 23 \cdot 43 \cdot 617$. The factors 13, 43, 617 are all NQR modulo 359. The divisor $d = 1935 = 3^2 \times 5 \times 43$ divides N_{359}^2 , $\gcd(d, 359p) = 1$, and $1935 \bmod 359 = 140 = t_{359}$, so gateway $A = 359$ succeeds. This claim is independently reproduced by `verify_hardest_prime.py` in the software repository.

This prime is the record-holder over the full 10^{11} range: the largest gateway parameter required for any prime $p \leq 10^{11}$. Of the four other primes requiring $A \geq 251$ (Theorem 6.2), one — $p = 6,372,847,849$ ($A = 311$) — also lies below 10^{10} , alongside the $A = 359$ record; the remaining three lie in $(10^{10}, 10^{11}]$. None exceeds $A = 359$.

Remark 6.3. The Case B QR7 residual class has relative density 2.14% at 10^{11} (down from 2.89% at 10^6), consistent with the relative-density-zero prediction for this class in [1]. The fraction requiring large A shrinks even faster: only 47 of the 88,088,687 Case B QR7 primes below 10^{11} need $A \geq 199$.

7. OPEN PROBLEMS

Question 7.1. Improve the bound to $E(X) = O(X/(\log X)^{3/2+\epsilon})$ for some $\epsilon > 0$ using properties of gateways beyond $A = 7$.

Question 7.2. Improve the exponent toward Conjecture 5.2 ($E(X) = O(X/(\log X)^{1+\delta})$ for all $\delta > 0$) by controlling the divisor character sums $\mathcal{N}_{A_i}(p)$ at gateways $A_i \neq 7$, not merely smoothness. As noted in §5, the joint Q_A -smoothness set does not contain the exceptional set, so the single-gateway smoothness method caps at the exponent $3/2$.

Question 7.3. Extend the verification beyond 10^{11} . The corrected pipeline (Section 6) confirms $A = 359$ as the hardest case for $p \leq 10^{11}$, unchanged since 10^{10} ; extending further would determine whether larger gateway parameters ever appear. The maximum A encountered for any prime $p \leq X$ grows slowly with X (log-like); bounding this growth rigorously is an open problem (the *bounded- A conjecture* of [1]).

Question 7.4. Characterise the set of primes p for which N_7 is Q_7 -smooth. By Theorem 1.1, this set has counting function $O(X/(\log X)^{3/2})$, but no explicit family is known. These are precisely the primes for which the $A=7$ gateway fails.

Question 7.5. Is there an $A \neq 7$ for which an exact characterisation of gateway failure holds for a *positive-density* subset of Case B QR7 primes? Theorem 4.3 rules out universal sharpness, but a density-one subset might still be achievable for specific A .

Question 7.6. Can the character sum formula (2) be combined with the large sieve to give an unconditional proof that $\mathcal{N}_A(p) > 0$ for all Case B QR7 primes $p \geq P_0$ (for explicit P_0)? This would require bounding $E_A(p)$ in terms of the main term and a zero-free region for $L(s, \chi)$ for characters $\chi \bmod A$.

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